THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Assertion (A): Energy flow in an ecosystem is unidirectional. Reason (R): Energy lost as heat cannot be reused by producers. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): The temperature in the stratosphere rises with height. Reason (R): The ozone layer absorbs ultraviolet rays from the Sun. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Seeds in cups: A (dry cotton), B (wet cotton, kept in light), C (wet cotton, kept in dark). Which pair isolates the effect of light? 1

(a) A, B and C together **(b)** A and C
(c) A and B **(d)** B and C

4. A student observes an onion peel with 10x eyepiece and 10x objective, and the cell length in the picture is 5 mm. The real length is: 1

(a) 0.5 mm **(b)** 0.05 mm **(c)** 0.005 mm **(d)** 5 mm

5. A town has 20,000 people. If 15,000 are vaccinated, the percentage vaccinated is: 1

(a) 15% **(b)** 50% **(c)** 25% **(d)** 75%

6. Four cells in a circuit are connected one way and the bulb glows. If one cell is reversed, the bulb: 1

(a) glows dimmer **(b)** does not change **(c)** melts **(d)** glows brighter

7. A man weighs 600 N on Earth. On a planet where gravity is 1/3 of Earth's, his weight is: 1

(a) 1800 N **(b)** 200 N **(c)** 600 N **(d)** 300 N

8. A box exerts 2000 Pa. If its base area is halved without changing weight, the pressure becomes: 1

(a) 2000 Pa **(b)** 500 Pa **(c)** 1000 Pa **(d)** 4000 Pa

9. A sample of a pure solid is heated at a constant rate and the temperature is recorded. A horizontal portion on the graph shows: 1

(a) chemical change **(b)** change of state **(c)** boiling only **(d)** freezing only

10. 9 g of water is decomposed. The masses of hydrogen and oxygen obtained are: 1

(a) 1 g and 8 g **(b)** 8 g and 1 g **(c)** 2 g and 7 g **(d)** 4.5 g and 4.5 g

11. A solution is made by mixing 200 g of 10 per cent solution and 300 g of 20 per cent solution of the same solute. The percentage of the final solution is: 1

(a) 16 per cent **(b)** 15 per cent **(c)** 14 per cent **(d)** 30 per cent

12. Two plane mirrors are at right angles. A ray hits the first at an angle of incidence of 25 degrees. The angle of incidence on the second is: 1

(a) 90 degrees **(b)** 65 degrees **(c)** 25 degrees **(d)** 50 degrees

13. A student measures the shadow of a 1 m stick as 0.5 m at noon on one day and 1 m a few months later. This shows the Sun's noon elevation has: 1

(a) stayed same **(b)** become zero **(c)** increased **(d)** decreased

14. Producers form the broad base of an ecosystem pyramid because they: 1

(a) do not need sunlight
(b) support all the other levels and hold the most energy
(c) are the largest animals
(d) eat the consumers

15. If the volume of water in oceans is about 1.35 billion cubic km and total fresh water is 3 per cent of total water, with total water about 1.4 billion cubic km, the fresh water volume is about: 1

(a) 1.4 million cubic km **(b)** 4.2 billion cubic km
(c) 420 thousand cubic km **(d)** 42 million cubic km

16. Which of these shows a good use of a control in a test on a new fertiliser? 1

(a) different plants different soil
(b) no plants
(c) all plants given fertiliser
(d) plants without fertiliser grown in same conditions

17. A dough is kept (i) in a warm place, (ii) in a fridge. Dough (i) rises more because: 1

(a) yeast is more active at warm temperature **(b)** yeast dies in warmth
(c) the fridge makes yeast grow faster **(d)** the cooler dough releases more CO2

18. An outbreak of dengue is reported in a colony. Which of the following steps is least useful? 1

(a) fogging by authorities **(b)** removing stagnant water
(c) boiling milk **(d)** using mosquito repellent

19. An electromagnet lifts a nail when connected to a 3 V cell. With a 6 V battery it lifts more nails because: 1

(a) iron becomes heavier **(b)** the coil is longer
(c) the current is larger **(d)** the nails are lighter

20. A magnet is brought near an iron nail and the nail jumps to the magnet. The force that does the work is: 1

(a) gravity **(b)** friction
(c) magnetic force **(d)** electrostatic force

21. A ship at sea is advised to move away from the path of a cyclone because: 1

(a) the eye is cold
(b) wind and waves are strongest near the eye wall
(c) ships are heavy
(d) the sea is calm

22. In a closed bottle half-filled with water at constant temperature, the rate of evaporation compared with the rate of condensation is: 1

(a) greater always **(b)** equal at equilibrium
(c) zero **(d)** less always

23. Zinc reacts with dilute acid and produces: 1

(a) hydrogen gas **(b)** chlorine gas **(c)** nitrogen gas **(d)** oxygen gas

24. Alloys such as brass are solutions in which: 1

(a) a liquid in a liquid **(b)** a gas in a liquid
(c) a solid dissolves in a solid **(d)** a gas in a gas

25. A torch bulb is placed at the focus of a concave reflector. The beam leaving the torch is: 1

(a) scattered in all directions **(b)** parallel
(c) divergent **(d)** convergent

26. A calendar with 12 months of 29.5 days is adjusted by adding a month every: 1

(a) every month **(b)** about 3 years **(c)** every year **(d)** every 10 years

27. The relationship between a tapeworm and its host is: 1

(a) predation **(b)** parasitism **(c)** competition **(d)** mutualism

28. Which is the most effective individual action to reduce carbon dioxide in the atmosphere? 1

(a) burning waste
(b) cutting trees
(c) using more fuel
(d) using public transport and saving electricity

29. Assertion (A): A hypothesis can be rejected by an experiment. Reason (R): A hypothesis must be a testable statement. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

30. Assertion (A): Antibiotics are not given for viral infections such as influenza. Reason (R): Viruses do not have the cell structures targeted by antibiotics. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

31. Assertion (A): Malaria can be controlled by draining stagnant water. Reason (R): Mosquito larvae develop in stagnant water. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

32. Assertion (A): A thick nichrome wire gets less hot than a thin nichrome wire when the same current passes through it. Reason (R): A thick wire has more resistance. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

33. Assertion (A): A body of mass 5 kg has the same mass on the Moon. Reason (R): Mass depends on the gravitational pull at that place. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

34. Assertion (A): A drinking straw can lift liquid into the mouth. Reason (R): Atmospheric pressure pushes the liquid up when pressure in the straw is reduced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

35. Assertion (A): The temperature stays constant while ice melts. Reason (R): The heat supplied is used to increase the kinetic energy of particles. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

36. Assertion (A): Sodium is stored under kerosene. Reason (R): Sodium is a very reactive metal. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

37. Assertion (A): Powdered sugar dissolves faster than sugar crystals. Reason (R): Powder has a smaller surface area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

38. Assertion (A): Convex mirrors are used as rear-view mirrors in vehicles. Reason (R): They always form real and enlarged images. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

39. Assertion (A): We always see the same side of the Moon. Reason (R): The Moon does not rotate on its axis. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

40. Assertion (A): Food chains rarely have more than four or five trophic levels. Reason (R): Most of the energy is lost as heat at each trophic level. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

41. Assertion (A): Venus is hotter than Mercury though it is farther from the Sun. Reason (R): Venus has a thick atmosphere rich in carbon dioxide that traps heat. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. A student tests four detergents but uses different amounts of water for each. The flaw is: 1

(a) wrong cloth colour **(b)** too many detergents
(c) stain is a quality **(d)** water amount is not controlled

43. Cells of a green leaf contain many chloroplasts because the leaf: 1

(a) absorbs minerals from the soil **(b)** gives support to the plant
(c) stores water for the plant **(d)** makes food by photosynthesis

44. Why is it risky if many people in a community skip vaccination? 1

(a) Water becomes impure
(b) Vaccines cause the disease
(c) Antibodies disappear
(d) The pathogen can spread through unvaccinated people

45. A student winds 50 turns on a nail and lifts 8 pins, and 100 turns and lifts 17 pins. The data show that: 1

(a) no relation **(b)** strength rises with turns
(c) strength falls with turns **(d)** pins have no mass

46. A 60 kg astronaut on the Moon has mass: 1

(a) 10 kg **(b)** 360 kg **(c)** 0 kg **(d)** 60 kg

47. A block of weight 60 N rests on a table. The force exerted by the table on it is: 1

(a) 30 N **(b)** 60 N upward **(c)** 60 N downward **(d)** 0

48. Ice at 0 degrees Celsius and water at 0 degrees Celsius are compared. Which statement is correct? 1

(a) Particles have the same kinetic energy but different arrangement
(b) Ice particles move faster
(c) Water has fewer particles
(d) Water has lower temperature

49. The total number of atoms in 2 molecules of H_2SO_4 is: 1

(a) 7 **(b)** 14 **(c)** 4 **(d)** 28

50. From 500 g of 20 per cent sugar solution, 100 g of water is evaporated. The new percentage is: 1

(a) 20 per cent **(b)** 30 per cent **(c)** 25 per cent **(d)** 10 per cent

END OF QUESTION PAPER **Total: 50 Marks**